

KD-real observables are generated by KD-positive pure states

Candidate full result for arXiv:2507.23628

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Status. Candidate full result, likely valid. We prove Conjecture (11) of the source for every second-countable locally compact abelian group. The separate conjecture about the convex hull of KD-positive states is not claimed.

1 Source conjecture

The source is Matéo Spriet, *Characterizing the Kirkwood–Dirac positivity on second countable LCA groups*, arXiv:2507.23628 (current PDF dated 23 February 2026) [1]. Its Theorem 1.3 proves two structural equalities for compact connected groups. The first is then conjectured to hold for every second-countable LCA (SCLCA) group. Figure 1 reproduces the exact passage.

where the former closure is taken with respect to the Hubert-Schmidt topology while the latter closed convex hull is taken in the trace norm topology.

The problem of characterizing groups for which one or two equalities of (10) hold remains open. We conjecture that

$$V_{\text{KDr}} = \overline{\text{span}_{\mathbb{R}}(\mathcal{E}_{\text{KD}+}^{\text{pure}})} \quad (11)$$

is always true. It is known to hold for any finite abelian group (10). Regarding the equality

$$\mathcal{E}_{\text{KD}+} = \text{conv}(\mathcal{E}_{\text{KD}+}^{\text{pure}}), \quad (12)$$

it is proved in (10) that it is true for the abelian groups $\mathbb{Z}/p^k\mathbb{Z}$, where p is any prime number and $k \geq 1$. However, the authors prove that it does not hold in general for finite abelian groups, as it does not hold for the groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/6\mathbb{Z}$. In view of those results and of the present paper, a lead for thought could be to consider SCLCA groups G for which the set of open and compact subgroups of G is totally ordered for inclusion. We conjecture that this property is sufficient for (12) to hold. It might also be necessary.

We conclude in section 4.5 by comparing some of our results with earlier

Figure 1: The two conjectures on page 7 of the official arXiv PDF.

Transcription of Conjecture (11). If V_{KDr} is the real vector space of self-adjoint Hilbert–Schmidt observables with real Kirkwood–Dirac distribution, and $\mathcal{E}_{\text{KD}+}^{\text{pure}}$ is the set of KD-positive pure states, then

$$V_{\text{KDr}} = \overline{\text{span}_{\mathbb{R}}(\mathcal{E}_{\text{KD}+}^{\text{pure}})}^{\|\cdot\|_2} \quad \text{for every SCLCA group } G. \quad (1.1)$$

2 Setup from the source

Fix dual Haar measures on G and $\widehat{G}$. Let

$$U(u, \eta) = M_\eta T_u, \quad \mathfrak{X}_A^0(u, \eta) = \text{Tr}(AU(u, \eta)).$$

The source proves that $A \mapsto \mathfrak{X}_A^0$ extends to a unitary map

$$\mathcal{T}_2(G) \longrightarrow L^2(G \times \widehat{G}). \quad (2.1)$$

Its Lemma 4.1 says that a self-adjoint $A \in \mathcal{T}_2(G)$ belongs to V_{KDr} exactly when $\mathfrak{X}_A^0$ is essentially supported on

$$N = \{(u, \eta) \in G \times \widehat{G} : \eta(u) = 1\}. \quad (2.2)$$

For a compact open subgroup $H \leq G$, write

$$H^\perp = \{\eta \in \widehat{G} : \eta(h) = 1 \ (h \in H)\}, \quad e_H = \mu_G(H)^{-1/2} \mathbf{1}_H,$$

and $P_H = |e_H\rangle\langle e_H|$. The source's Theorem 4.1 classifies all elements of $\mathcal{E}_{\text{KD}^+}^{\text{pure}}$: they are exactly

$$P_{H,a,\alpha} = U(a, \alpha) P_H U(a, \alpha)^*, \quad (2.3)$$

where H is compact open and $(a, \alpha) \in G/H \times \widehat{G}/H^\perp$.

3 Proof intuition

The support condition (2.2) converts the operator conjecture into a density problem on phase space. For each compact open H , the rectangle $H \times H^\perp$ lies in N . The characteristic function of P_H is its indicator. Weyl translation multiplies that indicator by a character, and the permitted translation parameters give *every* character of the compact group $H \times H^\perp$. Thus the pure states associated with one H already span the full L^2 space on that rectangle.

It remains to cover N . A fiber of N over u has positive Haar measure only when the cyclic closure of u is compact; the symmetric assertion holds for the η coordinate. For almost every $(u, \eta) \in N$, both cyclic closures are therefore compact. Starting with any compact open subgroup, one can enlarge a smaller compact open subgroup by the cyclic closure of u while staying inside $\ker \eta$. This puts (u, η) in some $H \times H^\perp$. Second countability reduces the cover to countably many rectangles, completing the density argument.

4 The phase-space covering

Call x in an LCA group *topologically periodic* when $\overline{\langle x \rangle}$ is compact.

Lemma 1 (rectangle cover). *Suppose G has a compact open subgroup. Then, up to a product-Haar-null set,*

$$N = \bigcup_{H \leq G \text{ compact open}} H \times H^\perp. \quad (4.1)$$

Moreover, a countable subfamily of the rectangles in (4.1) has the same union up to a null set.

Proof. For fixed $u \in G$, the vertical fiber of N is

$$N_u = \{\eta : \eta(u) = 1\} = \overline{\langle u \rangle}^\perp.$$

A closed subgroup of an LCA group has positive Haar measure exactly when it is open. By Pontryagin duality, $\overline{\langle u \rangle}^\perp$ is open exactly when $\overline{\langle u \rangle}$ is compact. Tonelli's theorem therefore shows that, for almost every $(u, \eta) \in N$, u is topologically periodic. Applying the same argument to the horizontal fibers shows that η is also topologically periodic almost everywhere.

Consider such a pair $(u, \eta) \in N$. Put $C = \overline{\langle u \rangle} \leq G$ and $D = \overline{\langle \eta \rangle} \leq \widehat{G}$; both are compact. Since $\eta(u) = 1$, continuity gives $C \subseteq \ker \eta$. Compactness of D implies that $D^\perp$ is open in G . Fix one compact open $K_0 \leq G$ and let

$$K = K_0 \cap D^\perp, \quad H = C + K.$$

Then K is compact open, so H is a compact subgroup containing the open subgroup K ; hence H is compact open. Both C and K lie in $\ker \eta$, and consequently $u \in H$ and $\eta \in H^\perp$. This proves (4.1) modulo the two null sets obtained from Tonelli.

Each $H \times H^\perp$ is open in the second-countable space $G \times \widehat{G}$. Their union is therefore an open Lindelöf subspace, so it has a countable subcover. Together with the null-set statement, this proves the final assertion. $\square$

5 Pure states give every local Fourier mode

Lemma 2 (Fourier basis on one rectangle). *For a compact open $H \leq G$,*

$$\mathfrak{X}_{P_{H,a,\alpha}}^0(u, \eta) = \mathbf{1}_H(u) \mathbf{1}_{H^\perp}(\eta) \eta(a) \overline{\alpha(u)}. \quad (5.1)$$

As (a, α) runs through $G/H \times \widehat{G}/H^\perp$, these functions form an orthonormal basis of $L^2(H \times H^\perp)$, extended by zero outside the rectangle.

Proof. Directly from the normalized indicator e_H ,

$$\begin{aligned} \mathfrak{X}_{P_H}^0(u, \eta) &= \langle e_H, M_\eta T_u e_H \rangle \\ &= \mathbf{1}_H(u) \mathbf{1}_{H^\perp}(\eta). \end{aligned} \quad (5.2)$$

Indeed, if $u \notin H$ the two cosets are disjoint, while for $u \in H$ the integral of η over H vanishes unless $\eta \in H^\perp$. Conjugating $M_\eta T_u$ by $U(a, \alpha) = M_\alpha T_a$ gives

$$U(a, \alpha)^* U(u, \eta) U(a, \alpha) = \eta(a) \overline{\alpha(u)} U(u, \eta),$$

which proves (5.1).

The dual of the compact group $H \times H^\perp$ is

$$\widehat{H} \times \widehat{H^\perp} \cong (\widehat{G}/H^\perp) \times (G/H).$$

Formula (5.1) realizes every character in this dual group. Fourier theory on compact abelian groups therefore makes these functions an orthogonal basis. Finally, $\mu_G(H) \mu_{\widehat{G}}(H^\perp) = 1$, so every basis vector has norm one. $\square$

6 Full resolution of Conjecture (11)

Theorem 3. *For every second-countable locally compact abelian group G ,*

$$V_{\text{KDr}} = \overline{\text{span}_{\mathbb{R}}(\mathcal{E}_{\text{KD}+}^{\text{pure}})}^{\|\cdot\|_2}.$$

Proof. If G has no compact open subgroup, Theorems 1.2 and 4.1 of [1] give $V_{\text{KDr}} = \{0\}$ and $\mathcal{E}_{\text{KD}+}^{\text{pure}} = \emptyset$, so the equality is immediate.

Assume that G has a compact open subgroup. Let $(H_n)_{n \geq 1}$ be the countable family from Lemma 1 and set $R_n = H_n \times H_n^\perp$. If $f \in L^2(N)$, partition the union into

$$E_1 = R_1, \quad E_n = R_n \setminus \bigcup_{j < n} R_j.$$

Then $f = \lim_{m \rightarrow \infty} \sum_{n \leq m} f \mathbf{1}_{E_n}$ in L^2 , and each summand belongs to $L^2(R_n)$. Lemma 2 shows that each such summand is an L^2 limit of complex linear combinations of characteristic functions of elements of $\mathcal{E}_{\text{KD}+}^{\text{pure}}$. Hence

$$L^2(N) = \overline{\text{span}_{\mathbb{C}}\{\mathfrak{X}_P^0 : P \in \mathcal{E}_{\text{KD}+}^{\text{pure}}\}}. \quad (6.1)$$

Now take $A \in V_{\text{KDr}}$. The source's Lemma 4.1 places $\mathfrak{X}_A^0$ in $L^2(N)$. By (2.1) and (6.1), there are finite complex combinations $B_m = \sum_j c_{m,j} P_{m,j}$, with $P_{m,j} \in \mathcal{E}_{\text{KD}+}^{\text{pure}}$, such that $B_m \rightarrow A$ in Hilbert–Schmidt norm. Since $A = A^*$ and every $P_{m,j}$ is self-adjoint,

$$C_m = \frac{B_m + B_m^*}{2} = \sum_j \text{Re}(c_{m,j}) P_{m,j}$$

belongs to $\text{span}_{\mathbb{R}}(\mathcal{E}_{\text{KD}+}^{\text{pure}})$ and

$$\|C_m - A\|_2 \leq \frac{1}{2}(\|B_m - A\|_2 + \|B_m^* - A\|_2) = \|B_m - A\|_2 \rightarrow 0.$$

This proves the nontrivial inclusion. The reverse inclusion follows because $\mathcal{E}_{\text{KD}+}^{\text{pure}} \subseteq V_{\text{KDr}}$ and V_{KDr} is a closed real subspace, as established in the source. $\square$

7 Verification, novelty, and boundary

The included verifier treats eight finite abelian groups, enumerates every subgroup H , builds every function in (5.1), and checks both the exact cover of N and the numerical rank of their span. In every case the rank equals $|N|$; this includes the noncyclic groups $(\mathbb{Z}/2)^2$ and $(\mathbb{Z}/2)^3$ and the composite cyclic group $\mathbb{Z}/6$. These tests agree with the finite-group theorem of De Bièvre–Langrenéz–Radchenko [2] and stress-test the phase factors and quotient parameters.

G	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_4$	$\mathbb{Z}_5$	$\mathbb{Z}_6$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2 \times \mathbb{Z}_3$	$\mathbb{Z}_2^3$
$ N $	3	5	8	9	15	10	15	36
computed rank	3	5	8	9	15	10	15	36

The proof's noncomputational checkpoints are: the Haar-positive-subgroup criterion, compact/open annihilator duality, the Tonelli null-set reduction, the compactness and openness of $C + K$, the Lindelöf subcover, and the full character parametrization in Lemma 2. No positivity claim is made for a general complex combination; positivity is unnecessary because Conjecture (11) concerns a real linear space.

Bounded searches of the run indexes, the exact conjecture, the source's later citations, and KD-real observable terminology through 13 August 2026 found no later proof for SCLCA groups. The

finite-group analogue is known [2]; the measure-theoretic rectangle cover is the new step claimed here. Novelty confidence is moderate-to-high, subject to expert literature review.

The distinct equality

$$\mathcal{E}_{\text{KD}+} = \overline{\text{conv}}(\mathcal{E}_{\text{KD}+}^{\text{pure}})$$

is not proved here. Attempts to extend the finite filtration decomposition to an infinite chain of compact open subgroups encounter an uncontrolled infinite positive-decomposition step. The present theorem settles exactly Conjecture (11), not Conjecture (12) or the proposed group characterization.

Human-review recommendation. Check the Tonelli reduction in Lemma 1, the annihilator identifications in Lemma 2, and the phase in (5.1). Then compare the rectangle-cover argument with terminology for topologically periodic elements in the LCA-group literature before asserting priority.

References

- [1] M. Spriet, *Characterizing the Kirkwood–Dirac positivity on second countable LCA groups*, arXiv:2507.23628 (2025; current PDF dated 23 February 2026).
- [2] S. De Bièvre, C. Langrenez, and D. Radchenko, *The Kirkwood–Dirac representation associated to the Fourier transform for finite abelian groups: positivity*, Ann. Henri Poincaré (2025), doi:10.1007/s00023-025-01614-7.